

$$I = \frac{dQ}{dt}; dI = \bar{j} dS; dQ = \rho dV = \rho dS dl$$

$$dI = \frac{dQ}{dt} = \rho v dS; j = \rho v; \bar{j} = q n \bar{v}$$

$$\rho = \frac{Q}{V} = \frac{Q_0}{V} = nq; I = \oint_S \bar{j} dS$$

$$\frac{dQ}{dt} = \oint_S \bar{j} \cdot d\vec{S}$$

$$\frac{d\rho}{dt} = \bar{\nabla} \cdot \bar{j}$$

$$\left(\oint_{\partial V} \bar{j} \cdot d\vec{S} = \lim_{\delta V \rightarrow 0} \frac{\oint_{\partial V} \bar{j} \cdot d\vec{S}}{\delta V} \right)$$

$$\bar{\nabla} \cdot \bar{j} = 0 - \text{сохранение заряда}$$

$$U = IR; U = E l; I = j S; R = \rho \frac{l}{S} \Rightarrow E = \rho j = \frac{j}{\sigma}$$